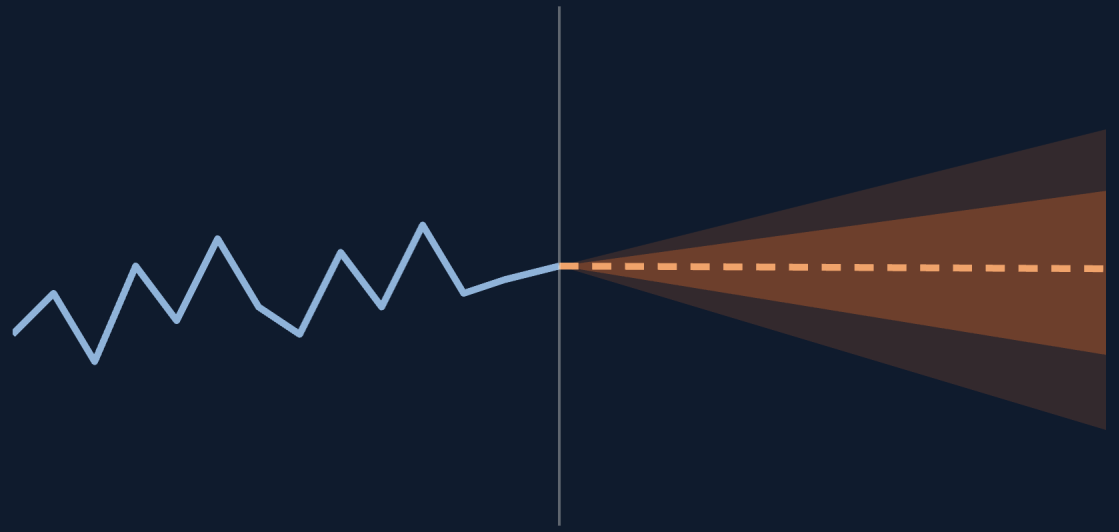
MACHINE LEARNING · PHASE 1

PRISM

Probabilistic Retail Inventory Forecasting with State-Space and Machine Learning Models



Local Kalman filtering vs. global gradient boosting on the M5 dataset

Newton School of Technology, Rishihood University

THE TEAM

Five members, one end-to-end pipeline

Team member	Enrollment No.	Primary module
Saumya Kumar	2401010432	Data pipeline, sampling, EDA
Krishiv	2401010234	Stockout marking, cold-start design
Shivansh Upadhayay	2401020109	Kalman filter and local baselines
Harsha Gonela	2401010181	Features, global LightGBM and PPT
Harshvardhan Gupta	2401010185	Backtest, metrics, significance tests, report

THE PROBLEM

Stocking decisions need the whole demand distribution, not just the mean

Intermittent

Most item-store series have long runs of zeros. The median series sells nothing on 63% of its active days.

Censored

A recorded zero may mean an empty shelf, not zero demand. M5 has no inventory data, so stockouts must be inferred.

Cold start

Assortments change, so many items have little or no history to learn from.

Research question: fit each SKU on its own (local), or one model across all SKUs (global)?

DATA

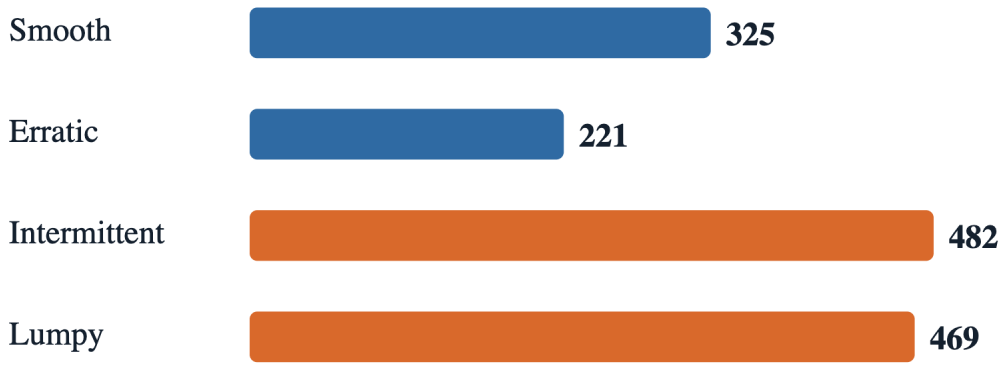
1,500 series from Walmart's M5 dataset

1,500
item-store series

3
stores: CA_1, TX_1, WI_1

- Weekends sell about 25% above the average day
- SNAP days lift FOODS sales by 6–16%
- Prices change in only about 1% of item-weeks
- Sample balanced by category and demand class, chosen with pre-origin data only

Syntetos–Boylan demand classes



63% of series are intermittent or lumpy

METHODS

Two modelling philosophies, one interface

LOCAL · ONE MODEL PER SKU

Structural Kalman filter

- Level + weekly seasonal state on log(1 + sales)
- Written from scratch; matches statsmodels to 10^{-9}
- Vectorised profile-likelihood MLE: all 1,500 SKUs in ~50 s
- Stockout days are skipped as missing observations

GLOBAL · ONE MODEL FOR ALL

LightGBM on lag features

- Horizon-safe direct design: lag-7 and lag-28 variants
- Lags, rolling stats, price, calendar, cross-store features
- Tweedie loss for the mean, pinball loss per quantile
- No-leakage property checked by a unit test

Baselines: **seasonal naive** and **TSB** (Croston-type). Every model outputs 7 quantiles (5%–95%) for 28 days ahead.

EVALUATION DESIGN

A fair, leakage-free comparison

Stockout marking

A zero-run is flagged when it is too long to be chance, judged only on past data. Flags are marked, never imputed: **13.7%** of days flagged.

Simulated cold start

150 items (10%) have their early history hidden, so they are seen with only **7–168 days** of data.

Rolling-origin backtest

5 origins × 28 days, refit at each origin. All models scored on the same 209,860 cells.

Metrics and tests

WAPE, RMSSE, scaled pinball loss (SPL), interval coverage; Diebold–Mariano and Wilcoxon tests.

RESULTS

Global LightGBM is best on every metric

~25%

lower WAPE than seasonal naive for both learned models

5.6%

lower SPL for LightGBM than the Kalman filter

5 / 5

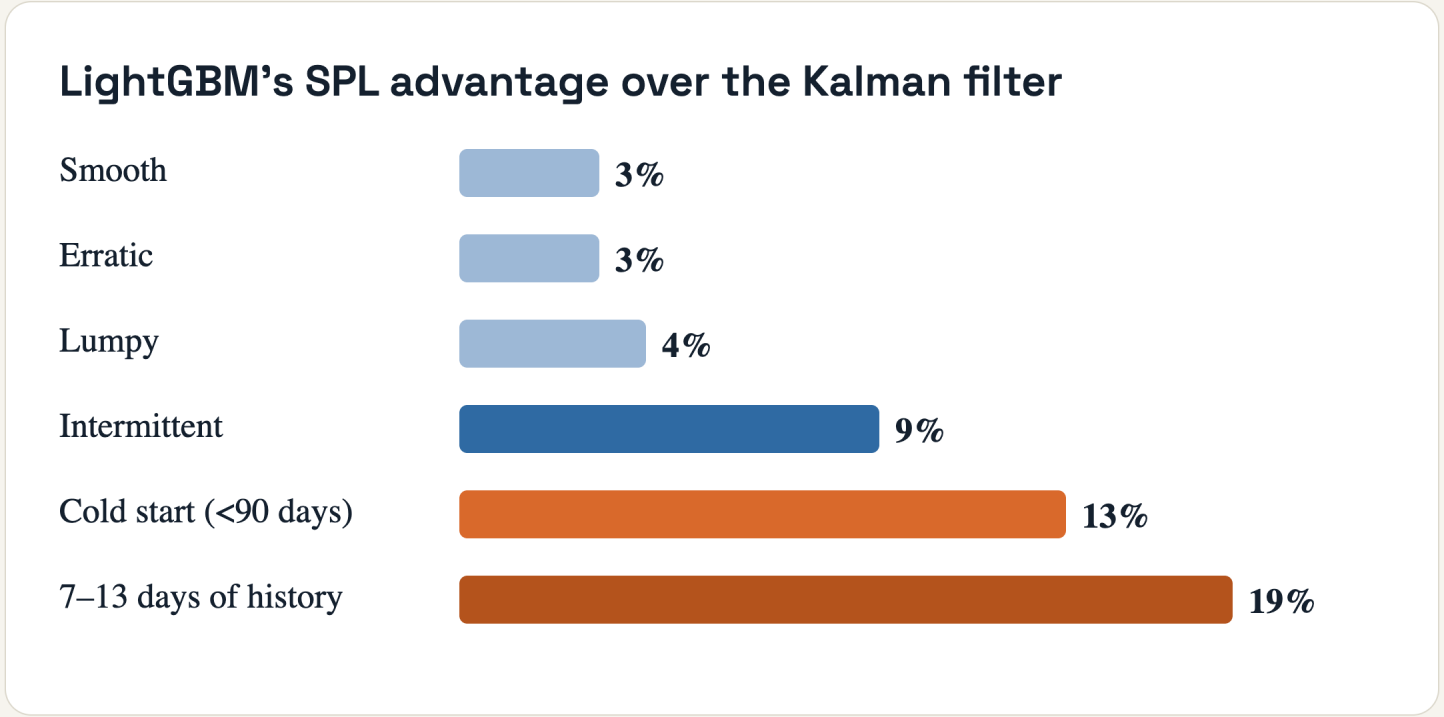
origins won by LightGBM; DM test $p < 10^{-20}$

Model (stockout-masked)	WAPE ↓	RMSSE ↓	SPL ↓	90% coverage
Seasonal naive	0.757	0.978	0.358	0.924
TSB (Croston-type)	0.600	0.753	0.281	0.887
Kalman filter (local)	0.582	0.744	0.280	0.903
LightGBM (global)	0.569	0.732	0.264	0.915

Source: rolling-origin backtest, 1,500 series

WHERE THE GAP GROWS

The sparser and newer the item, the more a global model helps



NEW ITEMS, 7–13 DAYS OF HISTORY

71%

of outcomes fall inside the Kalman filter's 90% interval, vs. 90% for LightGBM.

Too little data to estimate its own noise, so the local filter is over-confident.

Stockout-masked scaled pinball loss

FROM FORECAST TO ORDER

Order the quantile, not the point forecast

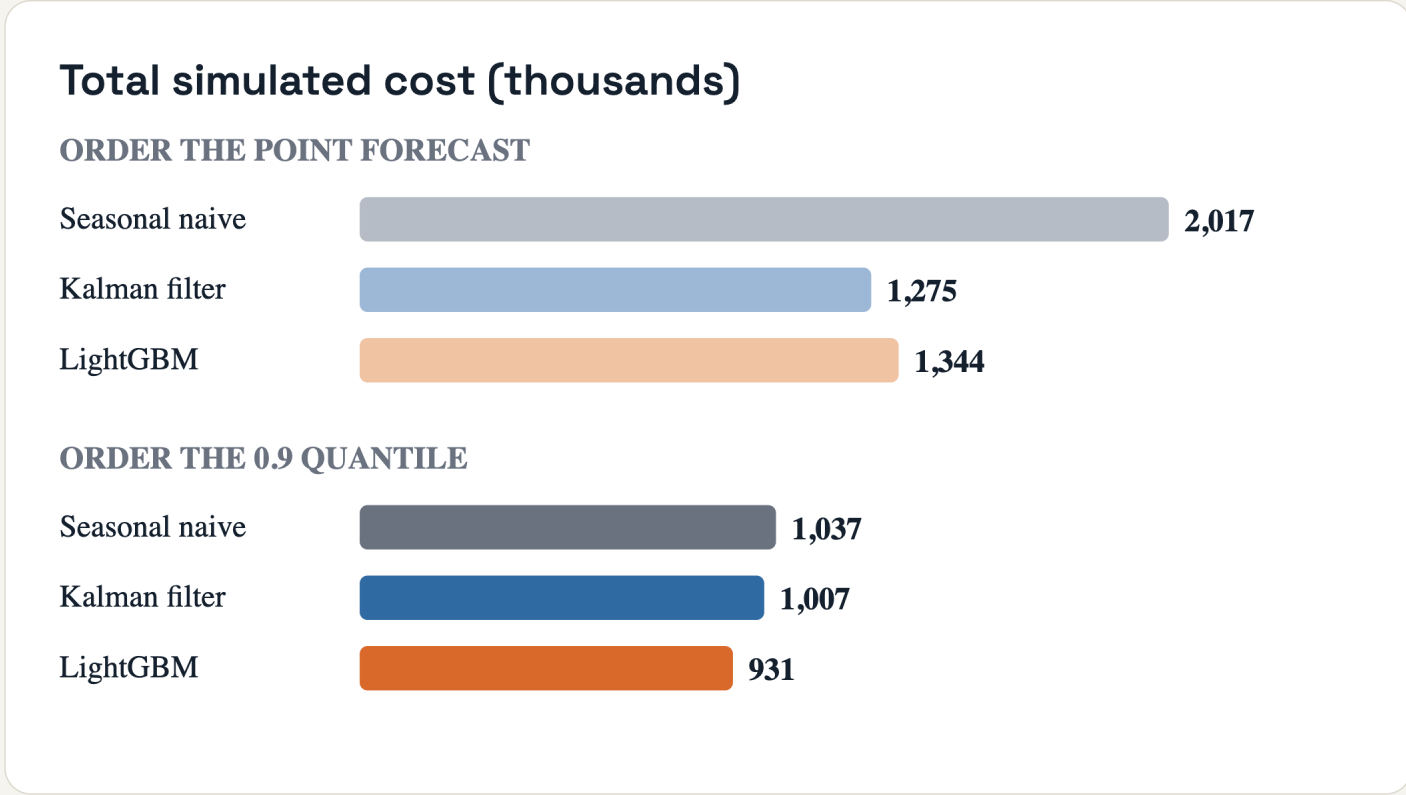
Newsvendor simulation: a lost sale costs **9×** an unsold unit, so the best order is the **0.9 quantile** of demand.

21–49%

lower inventory cost from ordering the quantile

7.5%

cheaper with LightGBM's quantiles than the filter's



Stockout-masked days, cu = 9, co = 1

CONCLUSIONS AND NEXT STEPS

Local filters fall short where data is thin

What we found

- Both learned models cut WAPE by about a quarter vs. naive
- Global LightGBM wins overall, most on intermittent and new items
- Better quantiles mean real savings in inventory cost

Phase 2

- Global **DeepAR** with a negative-binomial likelihood and item embeddings
- A **hybrid** that feeds Kalman level, season and variance into the network
- Same harness, metrics and segments

Thank you.